

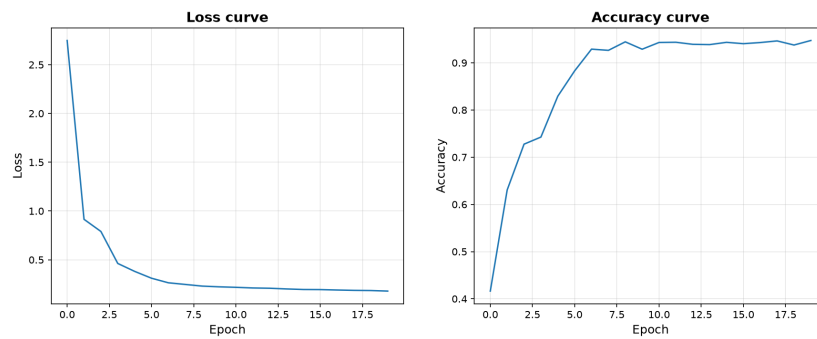
Model Performance Report: AI Text Detection

Course: Software Development Oriented to Machine Learning

Introduction

This report shows the results of our PyTorch Lightning model. We trained it to see if a text was written by a human or generated by an AI using data like word count and readability. Below are the graphs that show how the model worked in training. There was not a train-test split, so results may be extremely optimistic

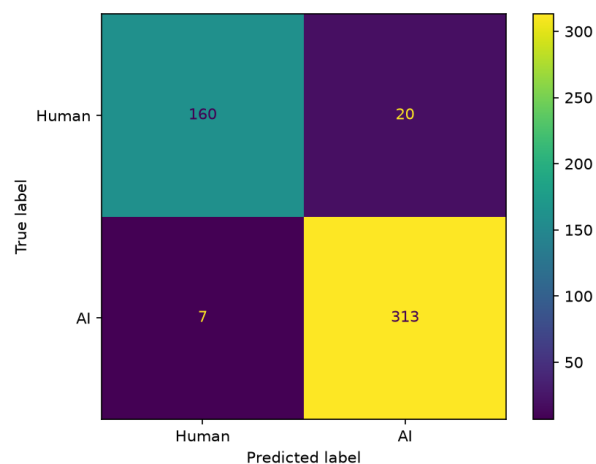
1. Training Progress (Loss & Accuracy)



Caption: This graph shows the training loss and accuracy for 20 epochs.

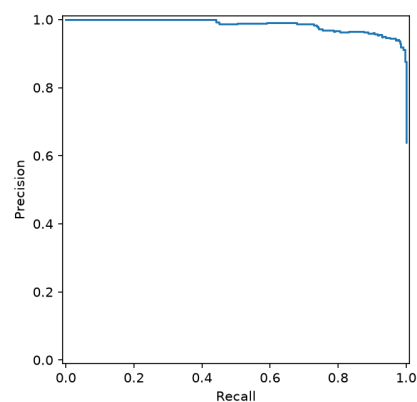
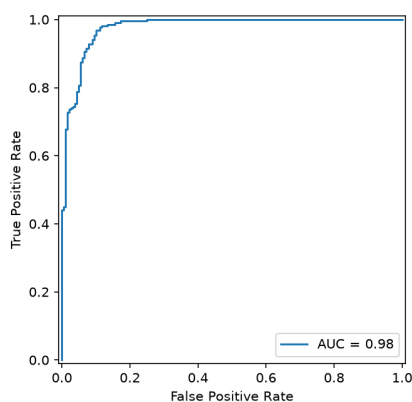
Interpretation: During the training phase, the model seemed to learn very well. The loss goes down and the accuracy goes up to around 95% at the end. However, the next graphs show that this success did not transfer to the final evaluation.

2. Confusion Matrix



Caption: This image shows the correct and incorrect predictions made by the model during the evaluation. **Interpretation:** The model did a very good job here. The main diagonal is full of examples. It correctly guessed around 160 Human texts and 300 AI texts, and it only made a couple false positives and some false negatives. This means the model completely understanding the nature of the data and how to separate each group.

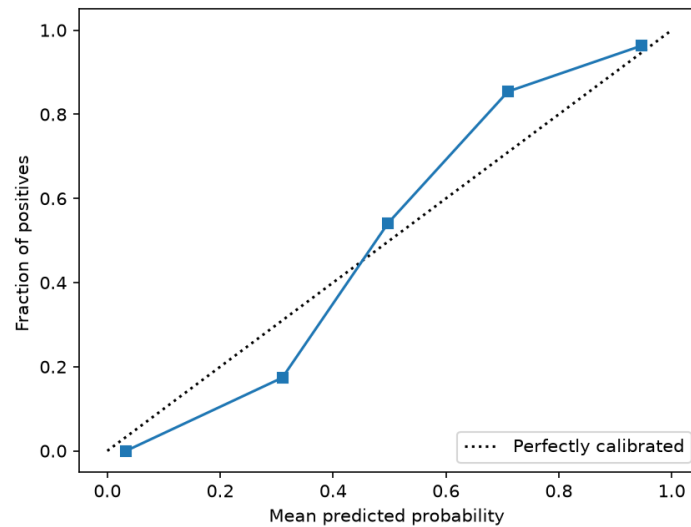
3. ROC and Precision-Recall Curves



Caption: These curves show how well the model separates the “Human” and “AI” classes. **Interpretation:** The ROC curve is almost perfect, it shapes very sharply around the left and top side, and the Area Under the Curve (AUC)

is almost the highest it can get. The Precision-Recall graph also proves the perfection. This tells us that the model has very good performance when trying to classify the documents and can clearly separate the two categories.

4. Model Calibration Analysis



Caption: This graph shows if the model's confidence matches reality. **Interpretation:** The dotted line is the “perfect” calibration. Our model's line is very close from this dotted line and overrides completely. This means we can completely trust the probabilities the model gives.